1) a)

$x = 3\cos 2t - 4\sin 2t$

$\dot{x} = \frac{\mathrm{d}x}{\mathrm{d}t} = -6\sin 2t - 8\cos 2t$

1) b)

$\ddot{x} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -12\cos 2t + 16\sin 2t$

1) c)

$\ddot{x} = -\omega^2 x$

$\Rightarrow -12\cos 2t + 16\sin 2t = -\omega^2(3\cos 2t - 4\sin 2t)$

$\Rightarrow -4(3\cos 2t - 4\sin 2t) = -\omega^2(3\cos 2t - 4\sin 2t)$

So $\ddot{x} = -4x$

$\Rightarrow \omega^2 = 4$

$\Rightarrow \omega = 2$ (ω is the angular frequency and is taken to be positive)

2) a)

$x = A\cos 3t + B\sin 3t$

$\dot{x} = \frac{\mathrm{d}x}{\mathrm{d}t} = -3A\sin 3t + 3B\cos 3t$

$\ddot{x} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -9A\cos 3t - 9B\sin 3t$

$\ddot{x} = -9(A\cos 3t + B\sin 3t) = -9x$

$\Rightarrow \omega^2 = 9$

$\Rightarrow \omega = 3$

2) b)

Substitute $x = 2$ and $t = 0$:

$2 = A\cos 0 + B\sin 0 \Rightarrow A = 2$

Substitute $\dot{x} = -3$ and $t = 0$:

$-3 = -3 \times 2 \times \sin 0 + 3B\cos 0$

$\Rightarrow -3 = 3B \Rightarrow B = -1$

So $A = 2$ and $B = -1$

3)

$\ddot{x} + \omega^2 x = 0$

Auxiliary Equation:

$m^2 + \omega^2 = 0$

$\Rightarrow m^2 = -\omega^2$

$\Rightarrow m = \pm\omega\mathrm{i}$

General Solution:

$x = A\cos\omega t + B\sin\omega t$

4)

$\ddot{x} + \omega^2 x = 0$

$x = A\cos\omega t + B\sin\omega t$

Substitute $x = x_0$ and $t = 0$:

$x_0 = A\cos 0 + B\sin 0 \Rightarrow A = x_0$

$\therefore x = x_0\cos\omega t + B\sin\omega t$

$\dot{x} = -\omega x_0\sin\omega t + B\omega\cos\omega t$

Substitute $\dot{x} = v_0$ and $t = 0$:

$v_0 = -\omega x_0\sin 0 + B\omega\cos 0$

$\Rightarrow v_0 = B\omega \Rightarrow B = \frac{v_0}{\omega}$

$\therefore x = x_0\cos\omega t + \frac{v_0}{\omega}\sin\omega t$

5) a)

$\ddot{x} + 36x = 0$

Auxiliary Equation:

$m^2 + 36 = 0$

$\Rightarrow m^2 = -36$

$\Rightarrow m = \pm 6\text{i}$

General Solution:

$x = A\cos 6t + B\sin 6t$

Substitute $x = 3$ and $t = 0$:

$3 = A\cos 0 + B\sin 0 \Rightarrow A = 3$

$\therefore x = 3\cos 6t + B\sin 6t$

Substitute $x = 4$ and $t = \frac{\pi}{12}$:

$4 = 3\cos\frac{\pi}{2} + B\sin\frac{\pi}{2}$

$\Rightarrow 4 = 0 + B \Rightarrow B = 4$

$\therefore x = 3\cos 6t + 4\sin 6t$

5) b)

$\ddot{x} + 400x = 0$

Auxiliary Equation:

$m^2 + 400 = 0$

$\Rightarrow m^2 = -400$

$\Rightarrow m = \pm 20\text{i}$

General Solution:

$x = A\cos 20t + B\sin 20t$

Substitute $x = -5$ and $t = 0$:

$-5 = A\cos 0 + B\sin 0 \Rightarrow A = -5$

$\therefore x = -5\cos 20t + B\sin 20t$

Substitute $x = 8$ and $t = \frac{3\pi}{40}$:

$8 = -5\cos\frac{3\pi}{2} + B\sin\frac{3\pi}{2}$

$\Rightarrow 8 = 0 - B \Rightarrow B = -8$

$\therefore x = -5\cos 20t - 8\sin 20t$

5) c)

$\ddot{x} + 2x = 0$

Auxiliary Equation:

$m^2 + 2 = 0$

$\Rightarrow m^2 = -2$

$\Rightarrow m = \pm\sqrt{2}\mathrm{i}$

General Solution:

$x = A\cos\sqrt{2}t + B\sin\sqrt{2}t$

Substitute $x = 1$ and $t = 0$:

$1 = A\cos 0 + B\sin 0 \Rightarrow A = 1$

$\therefore x = \cos\sqrt{2}t + B\sin\sqrt{2}t$

$\dot{x} = -\sqrt{2}\sin\sqrt{2}t + \sqrt{2}B\cos\sqrt{2}t$

Substitute $\dot{x} = \frac{1}{3}$ and $t = 0$:

$\frac{1}{3} = -\sqrt{2}\sin 0 + \sqrt{2}B\cos 0$

$\Rightarrow \sqrt{2}B = \frac{1}{3} \Rightarrow B = \frac{\sqrt{2}}{6}$

$\therefore x = \cos\sqrt{2}t + \frac{\sqrt{2}}{6}\sin\sqrt{2}t$

5) d)

$25\ddot{x} + x = 0$

Auxiliary Equation:

$25m^2 + 1 = 0$

$\Rightarrow m^2 = -\frac{1}{25}$

$\Rightarrow m = \pm\frac{1}{5}\mathrm{i}$

General Solution:

$x = A\cos\frac{1}{5}t + B\sin\frac{1}{5}t$

Substitute $x = \frac{1}{20}$ and $t = 0$:

$\frac{1}{20} = A\cos 0 + B\sin 0 \Rightarrow A = \frac{1}{20}$

$\therefore x = \frac{1}{20}\cos\frac{1}{5}t + B\sin\frac{1}{5}t$

$\dot{x} = -\frac{1}{100}\sin\frac{1}{5}t + \frac{1}{5}B\cos\frac{1}{5}t$

Substitute $\dot{x} = \frac{1}{100}$ and $t = 10\pi$:

$\frac{1}{100} = -\frac{1}{100}\sin 0 + \frac{1}{5}B\cos 2\pi$

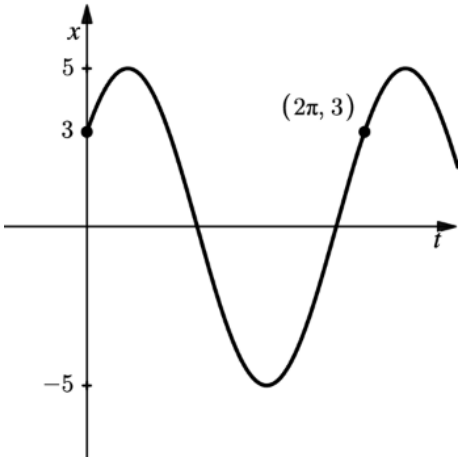
$\Rightarrow \frac{1}{5}B = \frac{1}{100} \Rightarrow B = \frac{1}{20}$

$\therefore x = \frac{1}{20}\cos\frac{1}{5}t + \frac{1}{20}\sin\frac{1}{5}t$

6) a)

$x = 3\cos t + 4\sin t$

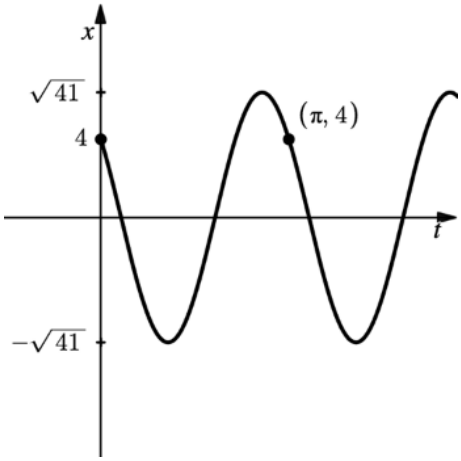
$\Rightarrow x = 5\sin\left(t + \arctan\frac{3}{4}\right)$



6) b)

$x = 4\cos 2t - 5\sin 2t$

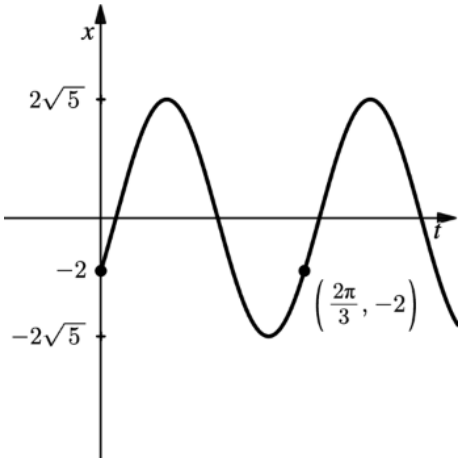
$\Rightarrow x = \sqrt{41}\cos\left(2t + \arctan\frac{5}{4}\right)$



6) c)

$x = 4\sin 3t - 2\cos 3t$

$\Rightarrow x = 2\sqrt{5}\sin(3t - \arctan 2)$



7) a)

$\ddot{x} + 9x = 0$

Auxiliary Equation:

$m^2 + 9 = 0$

$\Rightarrow m^2 = -9$

$\Rightarrow m = \pm 3\mathrm{i}$

General Solution:

$x = A\cos 3t + B\sin 3t$

Substitute $x = 2$ and $t = 0$:

$2 = A\cos 0 + B\sin 0 \Rightarrow A = 2$

$\therefore x = 2\cos 3t + B\sin 3t$

$\dot{x} = -6\sin 3t + 3B\cos 3t$

Substitute $\dot{x} = 6$ and $t = \frac{\pi}{3}$:

$6 = -6\sin \pi + 3B\cos \pi$

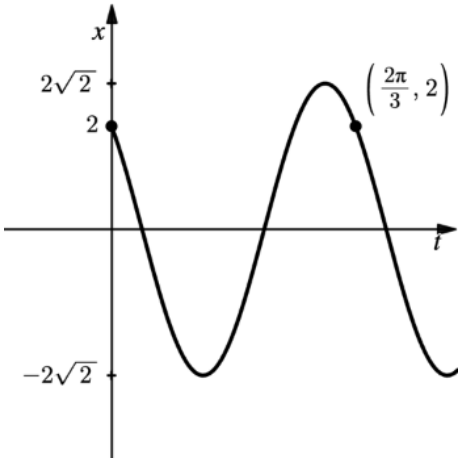
$\Rightarrow 6 = -3B \Rightarrow B = -2$

$\therefore x = 2\cos 3t - 2\sin 3t$

7) b)

$x = 2\cos 3t - 2\sin 3t$

$\Rightarrow x = 2\sqrt{2}\cos\left(3t + \frac{\pi}{4}\right)$



8) a)

$x\left(t+\frac{2\pi}{\omega}\right) = A\cos\left(\omega\left(t+\frac{2\pi}{\omega}\right)\right) + B\sin\left(\omega\left(t+\frac{2\pi}{\omega}\right)\right)$

$= A\cos(\omega t + 2\pi) + B\sin(\omega t + 2\pi)$

$= A(\cos\omega t\cos 2\pi - \sin\omega t\sin 2\pi) + B(\sin\omega t\cos 2\pi + \cos\omega t\sin 2\pi)$

$= A(\cos\omega t \times 1 - \sin\omega t \times 0) + B(\sin\omega t \times 1 + \cos\omega t \times 0)$

$= A\cos\omega t + B\sin\omega t = x(t)$ as required

8) b)

Period $= T = \frac{2\pi}{\omega}$

9)

$A\cos\omega t + B\sin\omega t \equiv R\cos(\omega t - \phi)$

$\equiv R(\cos\omega t\cos\phi + \sin\omega t\sin\phi)$

$\equiv R\cos\phi\cos\omega t + R\sin\phi\sin\omega t$

Compare Coefficients:

$A = R\cos\phi \Rightarrow \cos\phi = \frac{A}{R}$

$B = R\sin\phi \Rightarrow \sin\phi = \frac{B}{R}$

$R = \sqrt{A^2 + B^2}$

$\tan\phi = \frac{B}{A} \Rightarrow \phi = \arctan\frac{B}{A}$

$\therefore A\cos\omega t + B\sin\omega t \equiv \sqrt{A^2+B^2}\cos\left(\omega t - \arctan\frac{B}{A}\right)$

So $R = \sqrt{A^2+B^2}$ and $\phi = \arctan\frac{B}{A}$

10) a)

The acceleration $\ddot{x}$ is proportional to the displacement x with constant $-\omega^2$. Because of the negative sign, the acceleration will always be pointing in the opposite direction to the displacement.

10) b) i)

If $x > 0$ then $\ddot{x} = -\omega^2 x < 0$

10) b) ii)

If $x < 0$ then $\ddot{x} = -\omega^2 x > 0$

10) b) iii)

The acceleration always points towards $x = 0$, so $x = 0$ is an equilibrium point and the motion is "restoring" towards equilibrium.

11) a)

The acceleration $\ddot{x}$ is proportional to the displacement x with constant -7. Because of the negative sign, the acceleration will always be pointing in the opposite direction to the displacement.

11) b)

At equilibrium, acceleration must be zero.
Substitute $x = 0$:
$\ddot{x} = -7 \times 0 = 0$
So if the particle is at $x = 0$ and at rest, it will remain there.
Hence $x = 0$ is the equilibrium position.

11) c)

If the particle passes through $x = 0$ with positive velocity, it is moving in the positive direction. Immediately after passing through, $x > 0$ and so $\ddot{x} = -7x < 0$. The acceleration is negative, so the particle begins to slow down as it moves in the positive direction.

12) a)

Since $x > 0$, $\ddot{x} = -\omega^2 x < 0$
So the acceleration is negative

12) b)

If $x > 0$ and $\dot{x} < 0$, then the particle is moving in the negative direction back towards $x = 0$. So the particle is moving towards equilibrium.

12) c)

As $\dot{x} < 0$ and $\ddot{x} < 0$, the particle is accelerating in the negative direction, so the speed is increasing (until it hits its maximum speed at the equilibrium point).

13) a)

$x(t) = R\cos(\omega t - \phi)$

$\dot{x}(t) = -R\omega\sin(\omega t - \phi)$

13) b)

Speed $= |\dot{x}(t)| = |-R\omega\sin(\omega t - \phi)|$

Maximum speed $= R\omega$ as the maximum value of $\sin(\omega t - \phi)$ is 1

13) c)

Turning points occur when $\dot{x} = 0 \Rightarrow -R\omega\sin(\omega t - \phi) = 0$

$\Rightarrow \omega t - \phi = k\pi$ where k is an integer

If $\omega t - \phi = k\pi$ then:

$x = R\cos k\pi = \pm R$

So the displacement is $\pm R$ at the turning points.

$\therefore x = \pm R$ when $t = \frac{\phi + k\pi}{\omega}$ where k is an integer

14) a)

$x = R\cos(\omega t - \phi)$

Substitute $t = \frac{\phi}{\omega}$:

$x = R\cos\left(\omega \times \frac{\phi}{\omega} - \phi\right)$

$= R\cos(\phi - \phi)$

$= R\cos 0$

$= R$

The maximum displacement is R, so this occurs when $t = \frac{\phi}{\omega}$

14) b)

$x = R\cos(\omega t - \phi)$

Substitute $t = \frac{\phi}{\omega} + \frac{\pi}{2\omega}$:

$x = R\cos\left(\omega \times \left(\frac{\phi}{\omega} + \frac{\pi}{2\omega}\right) - \phi\right)$

$= R\cos\left(\phi + \frac{\pi}{2} - \phi\right)$

$= R\cos\frac{\pi}{2}$

$= R \times 0$

$= 0$

So the particle is at equilibrium when $t = \frac{\phi}{\omega} + \frac{\pi}{2\omega}$

14) c)

ϕ is known as the phase constant. It determines the initial position of the particle at $t = 0$ and where in the oscillation the motion begins. Graphically, it translates the graph horizontally.

15) a)

$x = R\cos\omega t$

$v = \dot{x} = -R\omega\sin\omega t$

15) b)

Speed $= |v| = |R\omega\sin\omega t|$

Maximum speed occurs when $|\sin\omega t| = 1$

So Maximum speed $= R\omega$

15) c)

At equilibrium, $x = 0$:

$0 = R\cos\omega t$

$\Rightarrow \cos\omega t = 0$

$\Rightarrow \sin\omega t = \pm 1$

At these points, the speed $= |v| = R\omega$

So the Maximum speed occurs at equilibrium

16) a)

$v = \frac{\mathrm{d}x}{\mathrm{d}t} = \dot{x}$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right) = \frac{\mathrm{d}}{\mathrm{d}t}(\dot{x}) = \ddot{x}$

So $\ddot{x} = \frac{\mathrm{d}v}{\mathrm{d}t}$

16) b)

Using the Chain Rule:

$\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}v}{\mathrm{d}x} \times \frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{d}v}{\mathrm{d}x} \times v = v\frac{\mathrm{d}v}{\mathrm{d}x}$

$\therefore \ddot{x} = v\frac{\mathrm{d}v}{\mathrm{d}x}$

17)

$\ddot{x} = -\omega^2 x$

$\Rightarrow v\frac{\mathrm{d}v}{\mathrm{d}x} = -\omega^2 x$

Use Separation of Variables:

$\Rightarrow \int v\,\mathrm{d}v = \int -\omega^2 x\,\mathrm{d}x$

$\Rightarrow \frac{1}{2}v^2 = -\frac{1}{2}\omega^2 x^2 + k$

$\Rightarrow v^2 + \omega^2 x^2 = c$

Where $c = 2k$

18) a)

At the furthest points (the turning points), the particle changes direction, so its instantaneous velocity is zero. Hence when $|x| = R \Rightarrow v = 0$

18) b)

$v^2 + \omega^2 x^2 = c$

Substitute $x = \pm R$ and $v = 0$:

$0^2 + \omega^2(\pm R)^2 = c$

$\Rightarrow c = \omega^2 R^2$

$\therefore v^2 + \omega^2 x^2 = \omega^2 R^2$

19) a)

$v^2 + 9x^2 = 36$

$\Rightarrow v^2 + 3^2x^2 = 6^2$

Since the relationship is of the form $v^2 + \omega^2x^2 = \omega^2R^2$ it corresponds to SHM.

19) b)

$\omega^2 = 3^2$

$\Rightarrow \omega = 3$

19) c)

$\omega^2R^2 = 36$

$\Rightarrow 9R^2 = 36$

$\Rightarrow R^2 = 4$

$\Rightarrow R = 2$

19) d)

Maximum speed occurs when $x = 0$ (at equilibrium)

$v^2 + 9 \times 0^2 = 36$

$\Rightarrow v^2 = 36$

$\Rightarrow v_{\max} = 6$ ms^{-1}

20) a)

$\ddot{x} + 25x = 0$

Auxiliary Equation:

$m^2 + 25 = 0$

$\Rightarrow m^2 = -25$

$\Rightarrow m = \pm 5\text{i}$

General Solution:

$x = A\cos 5t + B\sin 5t$

Substitute $x = 0.2$ and $t = 0$:

$0.2 = A\cos 0 + B\sin 0 \Rightarrow A = 0.2$

$\therefore x = 0.2\cos 5t + B\sin 5t$

$\dot{x} = -\sin 5t + 5B\cos 5t$

Substitute $\dot{x} = 0$ and $t = 0$:

$0 = -\sin 0 + 5B\cos 0 \Rightarrow B = 0$

So:

$x = 0.2\cos 5t$

20) b)

Period $= T = \frac{2\pi}{5}$

20) c)

$\dot{x} = -\sin 5t$

Maximum speed $= 1$ ms^{-1}

20) d)

$0 = 0.2\cos 5t$

$\Rightarrow \cos 5t = 0$

$\Rightarrow 5t = \frac{\pi}{2}$

$\Rightarrow t = \frac{\pi}{10}$ seconds

21) a)

$\ddot{x} + 36x = 0$

Auxiliary Equation:

$m^2 + 36 = 0$

$\Rightarrow m^2 = -36$

$\Rightarrow m = \pm 6\text{i}$

General Solution:

$x = A\cos 6t + B\sin 6t$

Period $= T = \frac{2\pi}{6} = \frac{\pi}{3}$

21) b)

Greatest Speed $= \omega R = 6 \times 0.5 = 3$ ms^{-1}

22) a)

$\ddot{x} + 4x = 0$

Auxiliary Equation:

$m^2 + 4 = 0$

$\Rightarrow m^2 = -4$

$\Rightarrow m = \pm 2\text{i}$

General Solution:

$x = A\cos 2t + B\sin 2t$

Period $= T = \frac{2\pi}{2} = \pi$

22) b)

Amplitude $= \frac{\omega R}{\omega} = \frac{8}{2} = 4$

23) a)

$\ddot{x} + \frac{1}{100}x = 0$

Auxiliary Equation:

$m^2 + \frac{1}{100} = 0$

$\Rightarrow m^2 = -\frac{1}{100}$

$\Rightarrow m = \pm\frac{1}{10}\mathrm{i}$

General Solution:

$x = A\cos\frac{1}{10}t + B\sin\frac{1}{10}t$

Angular Frequency $= \omega = \frac{1}{10}$

23) b)

$\ddot{x} = -\frac{1}{100}x$

$\Rightarrow v\frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{1}{100}x$

$\Rightarrow \int v\,\mathrm{d}v = \int -\frac{1}{100}x\,\mathrm{d}x$

$\Rightarrow \frac{1}{2}v^2 = -\frac{1}{200}x^2 + c$

When $x = 0.3$, $v = 0.04$:

$\frac{1}{2}\times 0.04^2 = -\frac{1}{200}\times 0.3^2 + c$

$\Rightarrow c = \frac{1}{800}$

$\therefore \frac{1}{2}v^2 = -\frac{1}{200}x^2 + \frac{1}{800}$

$\Rightarrow v^2 + \frac{1}{100}x^2 = \frac{1}{400}$

So:

$\omega^2 R^2 = \frac{1}{400}$

$\Rightarrow \frac{1}{100}R^2 = \frac{1}{400}$

$\Rightarrow R^2 = \frac{1}{4}$

$\Rightarrow R = \frac{1}{2}$

Amplitude $= 0.5$ m

23) c)

Period $= T = \frac{2\pi}{\frac{1}{10}} = 20\pi$

23) d)

Use:

$v^2 + \frac{1}{100}x^2 = \frac{1}{400}$

When $x = 0.4$:

$v^2 + \frac{1}{100} \times 0.4^2 = \frac{1}{400}$

$\Rightarrow v^2 = \frac{9}{10000}$

$\Rightarrow v = \frac{3}{100} = 0.03$ ms^{-1}

24) a)

Angular Frequency $= \omega = \frac{2\pi}{\frac{\pi}{2}} = 4$

24) b)

Use:

$v^2 + \omega^2 x^2 = \omega^2 R^2$

Substitute $v = 5$, $\omega = 4$ and $x = 0$:

$5^2 + 4^2 \times 0^2 = 4^2 R^2$

$\Rightarrow 16R^2 = 25$

$\Rightarrow R^2 = \frac{25}{16}$

$\Rightarrow R = \frac{5}{4}$

Amplitude $= 1.25$ m

24) c)

The particle starts from the centre of oscillation, so

$x = \frac{5}{4}\sin 4t$

Let $x = 1$:

$1 = \frac{5}{4}\sin 4t$

$\Rightarrow \sin 4t = \frac{4}{5}$

$\Rightarrow t = \frac{1}{4}\arcsin\frac{4}{5} = 0.232$ seconds (3 SF)

25) a)

At equilibrium, the acceleration is zero, so $\ddot{x} = 0$:

$\ddot{x} = -16x + 32$

$\Rightarrow 0 = -16(x - 2)$

So the equilibrium position is $x = 2$

25) b)

$\ddot{x} + 16x = 32$

Homogeneous Equation:

$\ddot{x} + 16x = 0$

Auxiliary Equation:

$m^2 + 16 = 0$

$\Rightarrow m^2 = -16$

$\Rightarrow m = \pm 4\text{i}$

Complementary Function:

$x = A\cos 4t + B\sin 4t$

Try this Particular Integral:

$x = k$

$\dot{x} = 0$

$\ddot{x} = 0$

$\ddot{x} + 16x = 32$

$\Rightarrow (0) + 16(k) = 32$

$\Rightarrow k = 2$

Particular Integral:

$x = 2$

General Solution:

$x = A\cos 4t + B\sin 4t + 2$

Substitute $x = 3$ and $t = 0$:

$3 = A\cos 0 + B\sin 0 + 2 \Rightarrow 3 = A + 2 \Rightarrow A = 1$

$\therefore x = \cos 4t + B\sin 4t + 2$

$\dot{x} = -4\sin 4t + 4B\cos 4t$

Substitute $\dot{x} = 0$ and $t = 0$:

$0 = -4\sin 0 + 4B\cos 0 \Rightarrow B = 0$

$\therefore x = \cos 4t + 2$

Amplitude $= R = 1$

25) c)

Period $= T = \frac{2\pi}{4} = \frac{\pi}{2}$ seconds

26) a)

At equilibrium, the acceleration is zero, so $\ddot{x} = 0$:

$\ddot{x} = -9x + 27$

$\Rightarrow 0 = -9(x - 3)$

So the equilibrium position is $x = 3$

26) b)

$\ddot{x} + 9x = 27$

Homogeneous Equation:

$\ddot{x} + 9x = 0$

Auxiliary Equation:

$m^2 + 9 = 0$

$\Rightarrow m^2 = -9$

$\Rightarrow m = \pm 3\mathrm{i}$

General Solution:

$x = A\cos 3t + B\sin 3t$

Try this Particular Integral:

$x = k$

$\dot{x} = 0$

$\ddot{x} = 0$

$\ddot{x} + 9x = 27$

$\Rightarrow (0) + 9(k) = 27$

$\Rightarrow k = 3$

Particular Integral:

$x = 3$

General Solution:

$x = A\cos 3t + B\sin 3t + 3$

So $\omega = 3$

26) c)

Substitute $x = 5$ and $t = 0$:

$5 = A\cos 0 + B\sin 0 + 3$

$\Rightarrow 5 = A + 3$

$\Rightarrow A = 2$

$\therefore x = 2\cos 3t + B\sin 3t + 3$

$\dot{x} = -6\sin 3t + 3B\cos 3t$

As the particle starts at $x = 5$ in the positive direction, and is moving back towards the equilibrium position at that time, the velocity must be negative.

Substitute $\dot{x} = -3$ and $t = 0$:

$-3 = -6\sin 0 + 3B\cos 0$

$\Rightarrow -3 = 3B$

$\Rightarrow B = -1$

$\therefore x = 2\cos 3t - \sin 3t + 3$

So $R = \sqrt{2^2 + (-1)^2} = \sqrt{5}$

26) d)

Maximum speed occurs at equilibrium:

$v_{\max} = \omega R = 3\sqrt{5}$ ms^{-1}

27)

Period $= T = 8\pi$

Amplitude $= R = 1.5$

Angular Frequency $= \omega = \frac{2\pi}{T} = \frac{2\pi}{8\pi} = \frac{1}{4}$

As the buoy passes upward through equilibrium at $t = 0$, we use

$x = 1.5 \sin \frac{1}{4} t$

This ensures $x = 0$ and the velocity is positive at $t = 0$

At A, $x = 0.5$:

$0.5 = 1.5 \sin \frac{1}{4} t$

$\Rightarrow \sin \frac{1}{4} t = \frac{1}{3}$

$\Rightarrow \frac{1}{4} t = \arcsin \frac{1}{3}$

$\Rightarrow t = 4 \arcsin \frac{1}{3}$

At B, $x = 1.25$:

$1.25 = 1.5 \sin \frac{1}{4} t$

$\Rightarrow \sin \frac{1}{4} t = \frac{5}{6}$

$\Rightarrow \frac{1}{4} t = \arcsin \frac{5}{6}$

$\Rightarrow t = 4 \arcsin \frac{5}{6}$

So the time taken for the buoy to travel from A to B is

$4 \arcsin \frac{5}{6} - 4 \arcsin \frac{1}{3} = 2.58$ seconds (3 SF)

28) a)

$v^2 + \omega^2 x^2 = \omega^2 R^2$

Substitute $x = 0.4$ and $v = 1.2$:

$1.2^2 + \omega^2 \times 0.4^2 = \omega^2 R^2$ (1)

Substitute $x = 0.8$ and $v = 0.6$:

$0.6^2 + \omega^2 \times 0.8^2 = \omega^2 R^2$ (2)

(2) – (1):

$0.6^2 - 1.2^2 + \omega^2(0.8^2 - 0.4^2) = 0$

$\Rightarrow 0.48\omega^2 = 1.08$

$\Rightarrow \omega^2 = \frac{9}{4}$

$\Rightarrow \omega = \frac{3}{2}$

Substitute into (1):

$1.2^2 + 1.5^2 \times 0.4^2 = 1.5^2 \times R^2$

$\Rightarrow R^2 = 0.8$

$\Rightarrow R = \frac{2\sqrt{5}}{5}$ metres

28) b)

Period $= T = \frac{2\pi}{\frac{3}{2}} = \frac{4\pi}{3}$ seconds

29) a)

$v^2 + \omega^2 x^2 = \omega^2 R^2$

Substitute $x = 15$ and $v = 120$:

$120^2 + \omega^2 \times 15^2 = \omega^2 R^2$ (1)

Substitute $x = 25$ and $v = 80$:

$80^2 + \omega^2 \times 25^2 = \omega^2 R^2$ (2)

(2) – (1):

$80^2 - 120^2 + \omega^2(25^2 - 15^2) = 0$

$\Rightarrow 400\omega^2 = 8000$

$\Rightarrow \omega^2 = 20$

$\Rightarrow \omega = 2\sqrt{5}$

Substitute into (1):

$120^2 + \left(2\sqrt{5}\right)^2 \times 15^2 = \left(2\sqrt{5}\right)^2 \times R^2$

$\Rightarrow R^2 = 945$

$\Rightarrow R = 3\sqrt{105}$ mm

29) b)

Period $= T = \frac{2\pi}{2\sqrt{5}} = \frac{\pi}{\sqrt{5}} = \frac{\sqrt{5}\pi}{5}$ seconds

30) a)

The force is towards O, so if x is the displacement from O then:

$F = -18x$

Newton's 2nd Law:

$F = ma \Rightarrow F = 2\ddot{x}$

$\Rightarrow 2\ddot{x} = -18x$

$\Rightarrow \ddot{x} = -9x$

30) b)

This is SHM as it is of the form $\ddot{x} = -\omega^2 x$ with $\omega^2 = 9 \Rightarrow \omega = 3$

30) c)

Period $= T = \frac{2\pi}{\omega} = \frac{2\pi}{3}$

31) a)

The force is towards O, so if x is the displacement from O then:

$F = -2x$

Newton's 2nd Law:

$F = ma \Rightarrow F = \frac{1}{2}\ddot{x}$

$\Rightarrow \frac{1}{2}\ddot{x} = -2x$

$\Rightarrow \ddot{x} = -4x$

31) b)

$\omega^2 = 4 \Rightarrow \omega = 2$

General Solution:

$x = A\cos 2t + B\sin 2t$

Substitute $t = 0$ and $x = 0.3$:

$0.3 = A\cos 0 + B\sin 0 \Rightarrow A = 0.3$

$\therefore x = 0.3\cos 2t + B\sin 2t$

$\dot{x} = -0.6\sin 2t + 2B\cos 2t$

Substitute $t = 0$ and $\dot{x} = 0$:

$0 = -0.6\sin 0 + 2B\cos 0$

$\Rightarrow 2B = 0 \Rightarrow B = 0$

$\therefore x = 0.3\cos 2t$

31) c)

First time at O is when $x = 0$:

$0 = 0.3\cos 2t$

$\Rightarrow \cos 2t = 0$

$\Rightarrow t = \frac{1}{2}\arccos 0 = \frac{1}{2} \times \frac{\pi}{2} = \frac{\pi}{4}$ seconds

31) d)

Use $v^2 + \omega^2 x^2 = \omega^2 R^2$

Substitute $\omega = 2$, $x = 0.2$ and $R = 0.3$:

$v^2 + 2^2 \times 0.2^2 = 2^2 \times 0.3^2$

$\Rightarrow v^2 = \frac{1}{5}$

$\Rightarrow v = \frac{\sqrt{5}}{5}$ ms^{-1}

32) a)

The force is towards O, so if x is the displacement from O then:

$F = -12x$

Newton's 2nd Law:

$F = ma \Rightarrow F = 0.8\ddot{x}$

$\Rightarrow \frac{4}{5}\ddot{x} = -12x$

$\Rightarrow \ddot{x} = -15x$

This is of the form $\ddot{x} = -\omega^2 x$, so the motion is SHM with $\omega^2 = 15 \Rightarrow \omega = \sqrt{15}$

32) b)

Period $= T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{15}}$

32) c)

$R = 0.25$

Maximum Speed $v_{\max} = \omega R = \sqrt{15} \times 0.25$

$= \frac{\sqrt{15}}{4}$ ms^{-1}

33) a)

The restoring force is:

$F = -8x$

Newton's 2nd Law:

$F = ma \Rightarrow F = 0.5\ddot{x}$

$\Rightarrow \frac{1}{2}\ddot{x} = -8x$

$\Rightarrow \ddot{x} = -16x$

This is of the form $\ddot{x} = -\omega^2 x$, so the motion is SHM with $\omega^2 = 16 \Rightarrow \omega = 4$

33) b)

At equilibrium $x = 0$, speed is at its maximum.

$v_{\max} = \omega R$

$\Rightarrow 1.2 = 4R$

$\Rightarrow R = 0.3$ metres

33) c)

The maximum extension equals the amplitude, so maximum extension $= 0.3$ metres

34) a)

$\ddot{x} = -9x$

General Solution:

$x = A\cos 3t + B\sin 3t$

Substitute $t = 0$ and $x = 0.5$:

$0.5 = A\cos 0 + B\sin 0 \Rightarrow A = 0.5$

$\therefore x = 0.5\cos 3t + B\sin 3t$

$\dot{x} = -1.5\sin 3t + 3B\cos 3t$

Substitute $t = 0$ and $\dot{x} = -1.5$ (as the box is moving towards equilibrium)

$-1.5 = -1.5\sin 0 + 3B\cos 0$

$\Rightarrow 3B = -1.5 \Rightarrow B = -0.5$

$\therefore x = 0.5\cos 3t - 0.5\sin 3t$

34) b)

Amplitude $= R = \sqrt{0.5^2 + (-0.5)^2} = \frac{\sqrt{2}}{2}$

34) c)

Maximum Speed

$v_{\max} = \omega R = 3 \times \frac{\sqrt{2}}{2}$

$= \frac{3\sqrt{2}}{2}$ ms^{-1}

35) a)

The restoring force is:

$F = -6x$

Newton's 2nd Law:

$F = ma \Rightarrow F = 1.2\ddot{x}$

$\Rightarrow 1.2\ddot{x} = -6x$

$\Rightarrow \ddot{x} = -5x$

35) b)

$\omega = \sqrt{5}$

General Solution:

$x = A\cos\sqrt{5}t + B\sin\sqrt{5}t$

Substitute $t = 0$ and $x = 0.6$:

$0.6 = A\cos 0 + B\sin 0 \Rightarrow A = 0.6$

$\therefore x = 0.6\cos\sqrt{5}t + B\sin\sqrt{5}t$

$\dot{x} = -0.6\sqrt{5}\sin\sqrt{5}t + B\sqrt{5}\cos\sqrt{5}t$

Substitute $t = 0$ and $\dot{x} = 0$:

$0 = -0.6\sqrt{5}\sin 0 + B\sqrt{5}\cos 0$

$\Rightarrow B\sqrt{5} = 0 \Rightarrow B = 0$

$\therefore x = 0.6\cos\sqrt{5}t$

35) c)

Equilibrium is when $x = 0$:

$0 = 0.6\cos\sqrt{5}t$

$\Rightarrow \cos\sqrt{5}t = 0$

$\Rightarrow t = \frac{1}{\sqrt{5}}\arccos 0 = \frac{1}{\sqrt{5}} \times \frac{\pi}{2} = \frac{\pi}{2\sqrt{5}} = \frac{\sqrt{5}\pi}{10}$ seconds

35) d)

3 assumptions:

The particle is modelled as a particle.

The motion is in a straight line.

There is no air resistance.

36) a)

The total restoring force is:

$F = -(5x + 5x) = -10x$

Newton's 2nd Law:

$F = ma \Rightarrow F = 1 \times \ddot{x}$

$\Rightarrow \ddot{x} = -10x$

This is of the form $\ddot{x} = -\omega^2 x$, so the motion is SHM with $\omega^2 = 10 \Rightarrow \omega = \sqrt{10}$

36) b)

Period $= T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{10}} = \frac{\sqrt{10}\pi}{5}$

36) c)

Use $v^2 + \omega^2 x^2 = \omega^2 R^2$

Substitute $\omega = \sqrt{10}$, $x = 0.2$ and $R = 0.4$:

$v^2 + \left(\sqrt{10}\right)^2 \times 0.2^2 = \left(\sqrt{10}\right)^2 \times 0.4^2$

$\Rightarrow v^2 = \frac{6}{5}$

$\Rightarrow v = \frac{\sqrt{30}}{5}$ ms^{-1}

37) a)

The total restoring force is:

$F = -(4x + 4x) = -8x$

Newton's 2nd Law:

$F = ma \Rightarrow F = 2 \times \ddot{x}$

$\Rightarrow 2\ddot{x} = -8x$

$\Rightarrow \ddot{x} = -4x$

This is of the form $\ddot{x} = -\omega^2 x$, so the motion is SHM with $\omega^2 = 4 \Rightarrow \omega = 2$

37) b)

Period $= T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi$

37) c)

x is the distance of the package from O towards B

So if the package is 1.1 m from A, the package's displacement is -0.4 m

Use $v^2 + \omega^2 x^2 = \omega^2 R^2$

Substitute $\omega = 2$, $x = -0.4$ and $R = 0.6$:

$v^2 + 2^2 \times (-0.4)^2 = 2^2 \times 0.6^2$

$\Rightarrow v^2 = \frac{4}{5}$

$\Rightarrow v = \frac{2\sqrt{5}}{5}$ ms^{-1}

38)

The total restoring force is:

$F = -(9mx + 9mx) = -18mx$

Newton's 2nd Law:

$F = ma \Rightarrow F = m\ddot{x}$

$\Rightarrow m\ddot{x} = -18mx$

$\Rightarrow \ddot{x} = -18x$

This is of the form $\ddot{x} = -\omega^2 x$, so the motion is SHM with $\omega^2 = 18 \Rightarrow \omega = 3\sqrt{2}$

The puck starts from O so when $t = 0$, $x = 0$. We also have that as the puck was tapped towards A with speed 0.8, we have that when $t = 0$, $\dot{x} = 0.8$

The greatest speed occurs when the particle is at equilibrium ($x = 0$), so the initial speed of 0.8 ms^{-1} is the maximum speed:

$v_{\max} = 0.8$

$\Rightarrow \omega R = 0.8$

$\Rightarrow 3\sqrt{2}R = 0.8$

$\Rightarrow R = \frac{2\sqrt{2}}{15}$

As $OA = 1$ m, the puck is closest to A a the turning point, so that distance $= 1 - \frac{2\sqrt{2}}{15} = 0.811$ m (3 SF)

39) a)

At equilibrium, the net force will be 0 N

Take positive direction as towards B

$\therefore -10mx + 5m(2.4 - x) = 0$

$\Rightarrow -2x + (2.4 - x) = 0$

$\Rightarrow 3x = 2.4$

$\Rightarrow x = 0.8$

So the equilibrium position is 0.8 m from A

39) b)

The total restoring force is:

$F = -10mx + 5m(2.4 - x)$

$= -10mx + 12m - 5mx$

$= 12m - 15mx$

Newton's 2nd Law:

$F = ma \Rightarrow F = m\ddot{x}$

$\Rightarrow m\ddot{x} = 12m - 15mx$

$\Rightarrow \ddot{x} = 12 - 15x$

$\Rightarrow \ddot{x} = -15(x - 0.8)$

This is SHM about $x = 0.8$ with $\omega^2 = 15 \Rightarrow \omega = \sqrt{15}$

39) c)

The right-hand spring force is $5m(2.4 - x)$

This is largest when $2.4 - x$ is largest, so when x is as small as possible during the motion.

We are told that the trolley is released from rest at a position 0.3 m to the right of its equilibrium position, so this is when $x = 0.8 + 0.3 = 1.1$ m

As the trolley is released from rest, this point is also the turning point, and the amplitude is $R = 0.3$

So the two extreme positions are $x = 0.8 \pm 0.3 = 1.1$ or 0.5

So the greatest right-spring force occurs at $x = 0.5$

$\therefore F = 5m(2.4 - 0.5) = \frac{19}{2}m$ N